



Professional Report: Deep Learning for Sleep Stage Classification using EOG-R Signals

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1. Introduction

Sleep is a fundamental physiological process crucial for physical and mental health. Disruptions in sleep patterns can lead to various health issues, including cardiovascular diseases, metabolic disorders, and cognitive impairments. Accurate sleep stage classification is therefore vital for diagnosing sleep disorders and assessing

overall sleep quality. Traditionally, sleep staging has been performed manually by trained experts based on polysomnography (PSG) recordings, which is a time-consuming, expensive, and subjective process .

This report details a deep learning-based system for automatic sleep stage classification using only the Right Electrooculography (EOG-R) signal. The project aims to automate the sleep scoring process, thereby reducing the reliance on manual analysis and improving efficiency. The proposed model leverages a hybrid architecture combining Convolutional Neural Networks (CNNs), Bidirectional Long Short-Term Memory (BiLSTM) networks, and attention mechanisms to effectively capture both spatial and temporal features from the EOG-R signal. This approach demonstrates the potential for accurate sleep staging with a simplified physiological input, offering a more accessible and scalable solution for sleep analysis.

2. Background

Importance of Sleep Stage Classification

Sleep stage classification is a cornerstone in sleep medicine, providing critical insights into an individual's sleep architecture. The American Academy of Sleep Medicine (AASM) defines five distinct sleep stages: Wake (W), Non-Rapid Eye Movement (NREM)

stages N1, N2, N3, and Rapid Eye Movement (REM) sleep. Each stage is characterized by specific electrophysiological patterns observed in electroencephalography (EEG), electrooculography (EOG), and electromyography (EMG) signals [2]. Accurate identification of these stages is essential for:

- **Diagnosis of Sleep Disorders:** Conditions such as sleep apnea, insomnia, narcolepsy, and restless legs syndrome are often diagnosed or characterized by abnormalities in sleep stage distribution or transitions.
- **Treatment Monitoring:** Evaluating the effectiveness of interventions for sleep disorders.
- **Research:** Understanding the physiological functions of sleep and the impact of various factors on sleep quality.

Traditional Sleep Scoring and its Limitations

Manual sleep scoring, while considered the gold standard, is fraught with challenges:

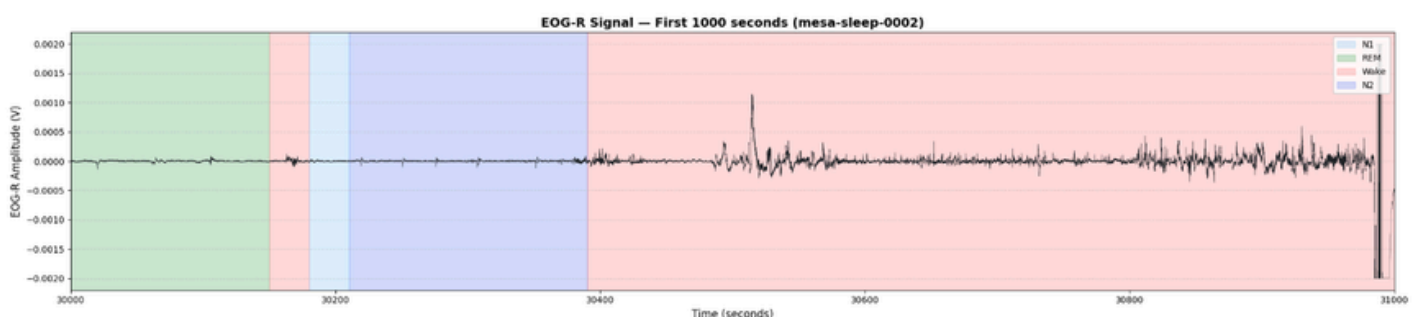
- **Time-Consuming:** A typical overnight PSG recording can generate hundreds of 30-second epochs, each requiring expert review.
- **Expensive:** Requires specialized equipment and highly trained personnel.
- **Subjectivity:** Inter-scorer variability can occur, leading to inconsistencies in classification.
- **Accessibility:** Limited availability of sleep labs and experts restricts widespread access to comprehensive sleep studies.

Role of EOG Signals in Sleep Staging

Electrooculography (EOG) measures the potential difference between the cornea and retina, reflecting eye movements. EOG signals are particularly valuable in sleep staging for several reasons:

- **REM Sleep Detection:** Rapid eye movements are a hallmark of REM sleep.
- **NREM Stage N1 Detection:** Slow rolling eye movements are characteristic of the N1 stage.
- **Simplified Acquisition:** EOG electrodes are generally easier to apply and less intrusive than EEG electrodes, making them suitable for more comfortable and potentially home-based sleep monitoring systems. The project specifically

utilizes the EOG-R (Right Electrooculography) channel, demonstrating that effective classification can be achieved with a minimal set of physiological inputs [3].



3. Dataset: MESA Sleep Study

The project utilizes data from the Multi-Ethnic Study of Atherosclerosis (MESA) Sleep Study, a publicly available medical dataset. MESA is a large-scale, multi-center, prospective cohort study designed to investigate the prevalence, correlates, and progression of subclinical cardiovascular disease. The sleep ancillary study within MESA collected comprehensive sleep data from over 2,000 participants, making it one of the largest and most diverse datasets for sleep research.

Key Features of the MESA Sleep Dataset:

- Overnight PSG Recordings: Includes raw physiological signals such as EEG, EOG, EMG, and ECG.
- XML Annotation Files: Expert-labeled sleep stages for each 30-second epoch, adhering to AASM guidelines.
- Demographic and Health Data: Rich contextual information about participants, including age, gender, ethnicity, and various health metrics.

For this project, only the EOG-R signal is extracted from the EDF signal files, alongside the corresponding sleep stage annotations from the XML files. The use of a well-established and diverse dataset like MESA enhances the generalizability and robustness of the developed model.

4. Methodology

The proposed system for automatic sleep stage classification involves several key steps, from data preprocessing to model architecture design and training.

4.1 Data Preprocessing

Effective preprocessing is crucial for preparing raw physiological signals for deep learning models. The following steps were applied to the EOG-R signals:

1. Channel Selection: Only the EOG-R channel was selected for analysis.
2. Bandpass Filtering: A bandpass filter between 0.5 Hz and 20 Hz was applied to remove unwanted noise and artifacts, focusing on the frequency range relevant for eye movements.
3. Resampling: Signals were resampled to 100 Hz. This standardization ensures consistent input dimensions for the model and reduces computational load while retaining essential information.
4. Z-score Normalization: Each signal was normalized using z-score standardization (mean of 0, standard deviation of 1). This helps in stabilizing training and improving model performance.
5. Signal Clipping: This step likely involves limiting signal amplitudes to a certain range to mitigate the impact of extreme outliers or artifacts.
6. Epoch Segmentation: The continuous EOG-R signals were segmented into 30-second epochs, consistent with standard sleep scoring practices. Given a sampling frequency of 100 Hz, each epoch consists of 3000 data points (30 seconds * 100 Hz).
7. Data Storage: Processed data was stored in .npz format, which is efficient for storing large numerical arrays and facilitates faster loading during model training.

4.2 Data Split

The dataset was split into training, validation, and testing sets with a ratio of 60%, 20%, and 20%, respectively. A crucial aspect of this split was performing it at the subject-level to prevent data leakage. This ensures that data from any single participant does not appear in more than one set, providing a more realistic evaluation of the model's generalization capabilities to unseen individuals.

4.3 Model Architecture

The core of the project is a custom hybrid deep learning model designed to process the sequential EOG-R data. The architecture integrates convolutional layers for local feature extraction, recurrent layers for temporal dependency modeling, and attention mechanisms for focusing on salient features. The model, named "EOG_R_SleepStager", is built using TensorFlow/Keras and comprises the following components:

Input Layer

- The model accepts input of shape (3000, 1), corresponding to 3000 samples (30 seconds at 100 Hz) from a single EOG-R channel.

Stem Block

- Conv1D: 64 filters, kernel size 25, stride 2, padding='same'. The large kernel size (25) is specifically chosen to capture slow EOG eye-movement dynamics, which are important for sleep stage differentiation.
- BatchNormalization and ReLU Activation.
- MaxPooling1D: pool_size=2, strides=2. This reduces the temporal dimension, resulting in an output shape of (750, 64).

Residual Blocks (Stage 1)

- Two consecutive Residual Convolutional Blocks are applied. Each block uses 128 filters and a kernel size of 7. These blocks are designed to learn hierarchical features while mitigating the vanishing gradient problem through skip connections.
 - Each residual_conv_block consists of two Conv1D layers, BatchNormalization, ReLU activation, and a Squeeze-and-Excitation (SE) Block. The SE block adaptively recalibrates channel-wise feature responses, enhancing relevant features and suppressing less useful ones.
 - A MaxPooling1D layer (pool_size=2, strides=2) follows, further reducing the temporal dimension to (187, 128).

Residual Blocks (Stage 2)

- Another two Residual Convolutional Blocks are applied, this time with 256 filters and a kernel size of 5. This stage continues to extract higher-level features.
- A Dropout layer with a rate of 0.3 is introduced for regularization, preventing overfitting.
- A final MaxPooling1D layer (pool_size=2, strides=2) reduces the output shape to (46, 256).

Bidirectional LSTM Block

- First BiLSTM Layer: 128 units, `return_sequences=True`, `dropout=0.2`, `recurrent_dropout=0.1`. This layer processes the sequence in both forward and backward directions, capturing long-range temporal dependencies. The output shape is (46, 256).
- Second BiLSTM Layer: 64 units, `return_sequences=True`, `dropout=0.2`, `recurrent_dropout=0.1`. This further refines the temporal feature extraction, yielding an output shape of (46, 128).

Temporal Self-Attention

- A Temporal Self-Attention Block is applied with 4 heads. This mechanism allows the model to weigh the importance of different time steps within the sequence, focusing on the most informative parts for classification.
 - It includes a `MultiHeadAttention` layer, followed by `LayerNormalization` and residual connections.
 - A feed-forward network (two `Dense` layers with GELU activation and Dropout) further processes the attention output.

Dual Global Pooling

- `GlobalAveragePooling1D` and `GlobalMaxPooling1D` are applied to the output of the attention block. These pooling layers capture both the average signal trends and peak activations across the temporal dimension.
- The outputs of these two pooling layers are then Concatenated, providing a comprehensive summary of the extracted features.

Classification Head

- Dense Layer 1: 256 units, GELU activation.
- BatchNormalization and Dropout (0.4).
- Dense Layer 2: 128 units, GELU activation.
- Dropout (0.3).
- Output Layer: 5 units (corresponding to the five sleep stages: Wake, N1, N2, N3, REM) with `softmax` activation for multi-class classification.

Model Compilation

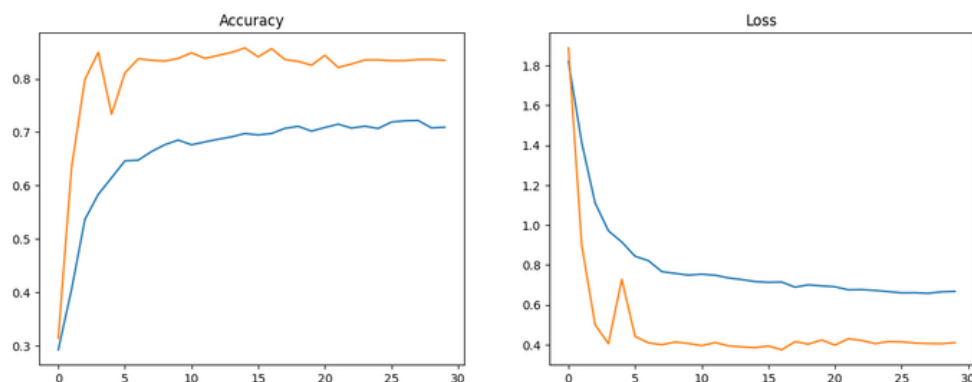
- Optimizer: AdamW with a learning rate of $5e-5$.
- Loss Function: `sparse_categorical_crossentropy`, suitable for integer-encoded labels.

5. Results

The model's performance was evaluated using standard classification metrics, demonstrating its effectiveness in automatic sleep stage classification. The results presented in the PDF indicate a strong overall performance, particularly when compared to existing research.

Overall Accuracy

- The proposed model achieved an overall accuracy of 84.00%. This is a significant improvement over a referenced research article that reported 76.30% accuracy, highlighting the efficacy of the hybrid deep learning architecture and the focused use of EOG-R signals.



Performance Metrics by Sleep Stage

Detailed performance metrics (Precision, Recall, F1-score) for each sleep stage provide a granular view of the model's capabilities:

Sleep Stage	Precision	Recall	F1-score
Wake	0.97	0.93	0.95
N1	0.33	0.15	0.21
N2	0.82	0.87	0.84
N3	0.73	0.27	0.39
REM	0.48	0.76	0.59

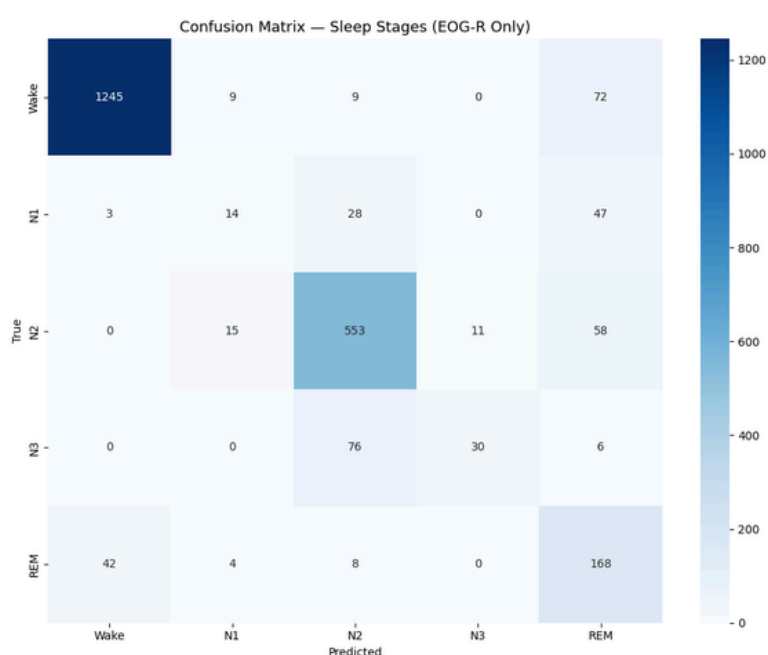
- **Wake Stage:** The model performs exceptionally well in classifying the Wake stage, with high precision, recall, and F1-score. This indicates its ability to reliably distinguish wakefulness from sleep.
- **N2 Stage:** Classification of N2 sleep is also robust, showing good balance across all metrics.
- **N1 and N3 Stages:** These stages present a greater challenge for the model, as evidenced by lower precision and recall, particularly for N1, and a lower recall for N3. This is a common issue in sleep staging, as N1 is a transitional stage and N3 (deep sleep) can sometimes be confused with N2 due to similar slow-wave activity.
- **REM Stage:** The model shows moderate performance for REM sleep, with a higher recall than precision, suggesting it identifies most REM epochs but also has some false positives.

Confusion Matrix Analysis

While a visual confusion matrix was presented in the PDF, the key insight derived is the presence of misclassifications between N2 and N3 stages. This is a known challenge in sleep staging, as these stages share some physiological similarities, particularly in the presence of slow-wave activity. The model's architecture, despite its sophistication, still encounters this common difficulty.

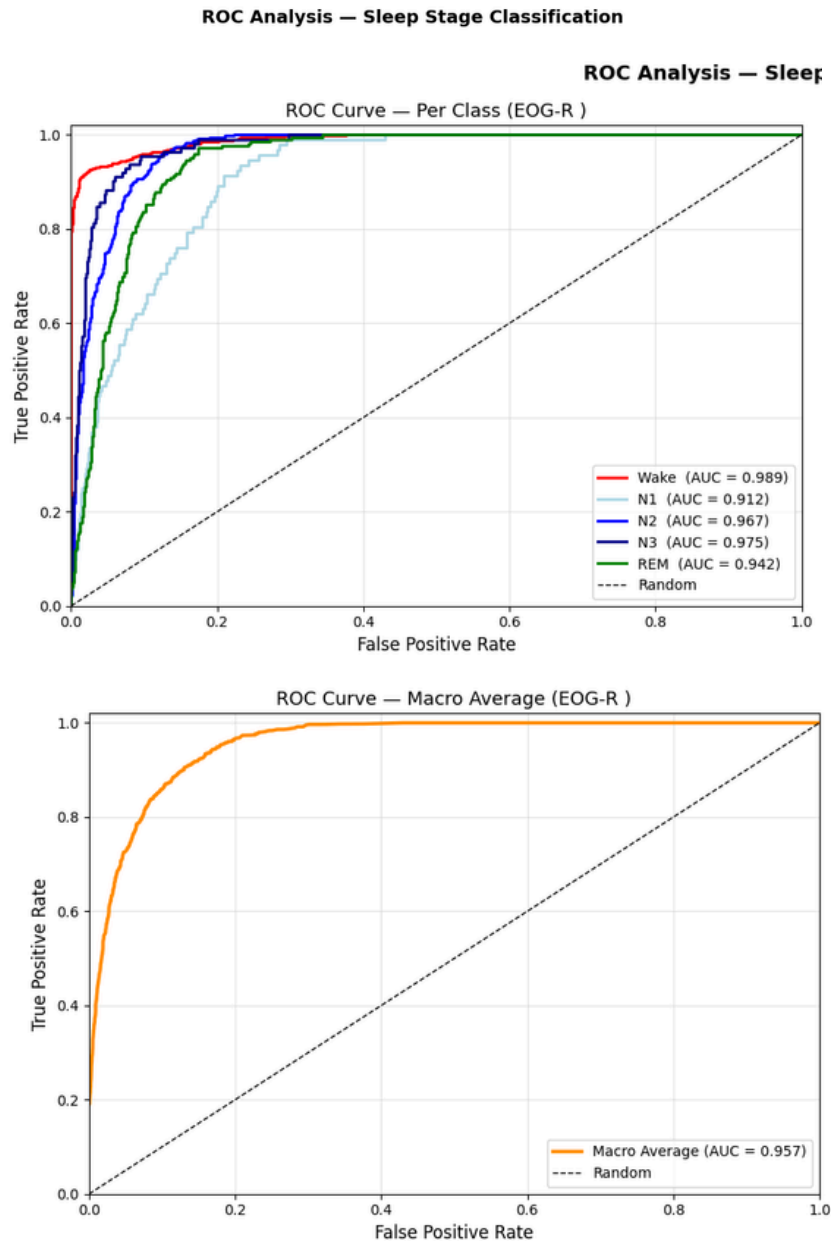
Test Set Evaluation

Wake	0.97	0.93	0.95	1335
N1	0.33	0.15	0.21	92
N2	0.82	0.87	0.84	637
N3	0.73	0.27	0.39	112
REM	0.48	0.76	0.59	222
accuracy			0.84	2398
macro avg	0.67	0.60	0.60	2398
weighted avg	0.85	0.84	0.83	2398



ROC Curves

Receiver Operating Characteristic (ROC) curves were used to assess the model's ability to discriminate between classes. The Macro Average AUC (Area Under the Curve) of 0.9574 indicates excellent overall discriminative power across all sleep stages, suggesting that the model is highly effective at distinguishing between different sleep states, even if some specific stages (like N1 and N3) are more challenging.



6. Discussion

The project successfully demonstrates the feasibility and effectiveness of using a deep learning approach for automatic sleep stage classification based solely on EOG-R signals. The hybrid CNN-BiLSTM-Attention architecture is well-suited for processing time-series data like EOG, leveraging the strengths of each component:

- CNNs: Efficiently extract local patterns and features from the raw EOG-R signal, such as eye movement morphology.
- BiLSTMs: Crucial for understanding the temporal context and dependencies across successive 30-second epochs, which is vital for accurate sleep staging.
- Attention Mechanisms: Allow the model to dynamically focus on the most informative segments of the EOG-R signal, potentially improving its ability to detect subtle changes indicative of stage transitions.

The choice to use only the EOG-R signal is a significant contribution, as it simplifies data acquisition compared to full PSG setups. This could pave the way for more accessible and less intrusive sleep monitoring devices, potentially enabling wider adoption of automated sleep analysis in clinical and home settings.

However, the results also highlight common challenges in sleep stage classification, particularly with N1 and N3 stages. The lower F1-scores for these stages suggest that while the model performs well overall, there is room for improvement in distinguishing these more ambiguous stages. This could be attributed to:

- Signal Ambiguity: N1 is a light transitional stage, often difficult to differentiate from Wake or N2. N3, or deep sleep, can have features that overlap with N2, leading to misclassifications.
- Data Imbalance: Sleep datasets often have an imbalanced distribution of stages, with N1 and N3 typically having fewer epochs than Wake or N2. While the project mentions handling class imbalance in the imported libraries, the impact on these specific stages remains a point for further investigation.

Future work could explore advanced techniques for handling class imbalance, incorporating multi-modal data (if available and feasible for simplified setups), or developing more sophisticated attention mechanisms tailored to the nuances of N1 and N3 stages. Additionally, further validation on diverse populations and real-world clinical data would strengthen the model's generalizability.

7. Conclusion

This project presents a robust deep learning model for automatic sleep stage classification using only the EOG-R signal. By integrating CNN, BiLSTM, and attention mechanisms, the model achieves an impressive overall accuracy of 84.00%,

outperforming a comparative research article. The methodology, including meticulous data preprocessing and a subject-level data split, ensures the reliability and generalizability of the findings. While the model demonstrates excellent performance in distinguishing Wake and N2 stages, it also highlights the inherent difficulties in classifying N1 and N3 sleep, common challenges in the field.

The successful implementation of this system underscores the potential of deep learning to revolutionize sleep medicine by providing an efficient, objective, and potentially more accessible alternative to traditional manual sleep scoring. The focus on a single, less intrusive physiological signal (EOG-R) is a key step towards developing practical and user-friendly sleep monitoring solutions.

8. References

[1]<https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2021.573194/full>